

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.

- | | | |
|---|-----------|---|
| A) Differentiate Between Biological Neural Networks and Artificial Neural Networks. | (BT2/CO1) | 6 |
| B) Define Neuron, Bias, Weights, activation function, loss function, and Optimizer. | (BT1/CO1) | 6 |
| C) What is perceptron? Define Neural Network architecture for implementing AND Gate | (BT3/CO1) | 6 |

Q.2 Solve Any Two of the following.

- | | | |
|---|-----------|---|
| A) What is the Forward Propagation? Explain with a suitable example. | (BT3/CO2) | 6 |
| B) What is the significance of the activation function? Explain any four activation functions with formula and use. | (BT3/CO2) | 6 |
| C) What is an optimizer? Explain the Gradient descent algorithm in detail. | (BT3/CO2) | 6 |

Q. 3 Solve Any Two of the following.

- | | | |
|--|-----------|---|
| A) What is a computation graph? Explain the role of derivatives in ANN. | (BT3/CO3) | 6 |
| B) Define backpropagation. Describe the basic steps in backpropagation learning. | (BT2/CO3) | 6 |
| C) Explain the feedforward Neural Network with a suitable example. | (BT3/CO1) | 6 |

Q.4 Solve Any Two of the following.

- | | | |
|--|-----------|---|
| A) Discuss various techniques to deal with overfitting in ANN. | (BT3/CO4) | 6 |
| B) What are different preprocessing techniques used for the generalization performance of ANN? | (BT2/CO4) | 6 |
| C) Discuss Bias and Various tradeoffs. | (BT3/CO4) | 6 |

Q. 5 Solve Any Two of the following.

- | | | |
|---|-----------|---|
| A) Differentiate between fuzzy set and Boolean set? Explain any four operations on a fuzzy set. | (BT3/CO5) | 6 |
|---|-----------|---|

- | | | | |
|-----------|---|-----------|----------|
| B) | Explain the basic architecture of the fuzzy system. | (BT4/CO5) | 6 |
| C) | What is a Genetic Algorithm? Explain any four applications of the Genetic Algorithm | (BT3/CO5) | 6 |

***** End *****